

PANGANI GIRLS' KCSE PREDICTION



EXAM TEST-2025

231/1

BIOLOGY

THEORY

PAPER 1

TIME: 2 HOURS

NAME.....

SCHOOL.....

SIGN.....

INDEX NO

DATE.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- 1) Write your name and Index Number in the spaces provided above.
- 2) Sign and write date of examination in the spaces provided above.
- 3) Answer **ALL** questions in the spaces provided.
- 4) All workings **MUST** be clearly shown where necessary.

FOR EXAMINERS USE ONLY.

Question	Maximum Score	Candidates Score
1 – 25	80	

Answer ALL questions in the spaces provided.

1. Explain the **term** Binomial Nomenclature.

(1mk)

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2. Name **three** forces involved in transportation of water and mineral salts.

(3mks)

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3. (a) Give **two** roles of DNA.

(2mks)

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(b) State the difference between **DNA** and **RNA**.

(1mk)

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4. Two strips A and B were cut from Tradescantia whose cell sap was 30% sugar. Strip A was placed in a solution of 10% sugar concentration while strip B was placed in 50% sugar concentration.

(a) What change was expected in strips **A** and **B**?

(2mks)

Strip **A**:

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Strip **B**:

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(b) Account for the results in strip A. (3mks)

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5. State the biological significance of each of the following:

(a) Thick muscular walls and narrow lumen in arteries. (1mk)

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(b) Narrow xylem vessels in flowering plants. (1mk)

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6. Suggest three reasons why green plants are included in a fish aquarium. (3mks)

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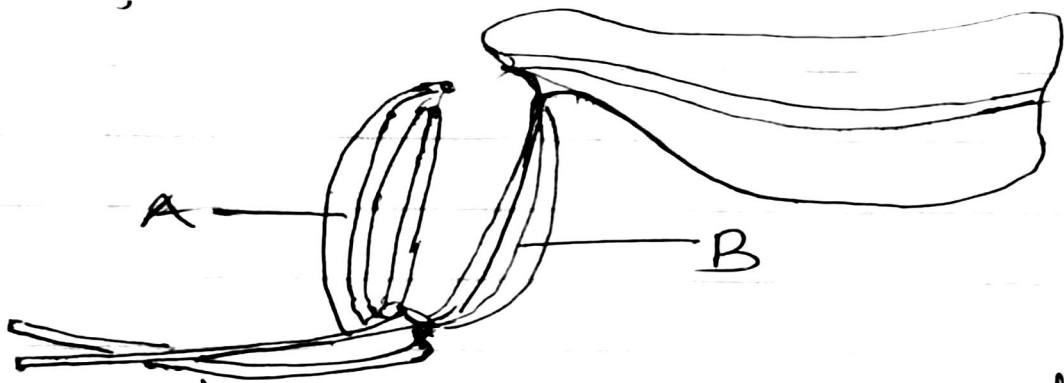
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7. (a) Study the diagram below and answer the questions that follow.



- Name the muscle labelled: **(2mks)**
A:.....
B:.....

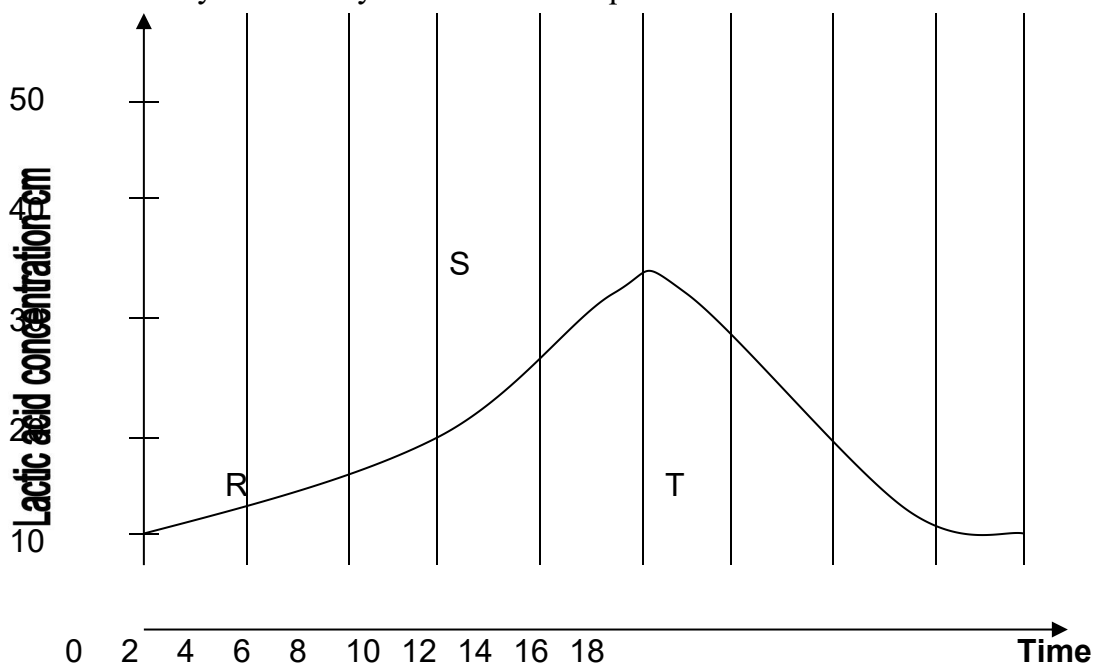
- What happens to each muscle as the arm is straightened? **(2mks)**
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8. The binomial name of housefly is MUSCA DOMESTICA.

- State **two** mistakes in the way the scientific name is written. **(2mks)**
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- Re-write the name in correct manner following the rules of binomial nomenclature.**(1mk)**
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9. The diagram below shows the general appearance of lactic acid in the blood of an athlete after an exercise. Study it carefully and answer the questions that follow:



- (a) Name the physiological process represented by the above diagram. **(1mk)**

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(b) Explain what happened in the body between points:

(i) R and S **(1mk)**

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(ii) S and T **(1mk)**

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10. State the use of each of the following apparatus:

- Bait trap **(1mk)**

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- Specimen bottle **(1mk)**

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- Pitfall trap **(1mk)**

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11. **(a)** Define the term organic evolution. **(1mk)**

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(b) Give two examples of vestigial structures. **(2mks)**

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12. (a) Distinguish between epigeal and hypogeal germination. (1mk)

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(b) Why is oxygen necessary in the germination of seeds? (2mks)

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13. (a) Digestion in the stomach involves the gastric juice, which contains mucus as one of its components. State the role of mucus in the digestion process. (1mk)

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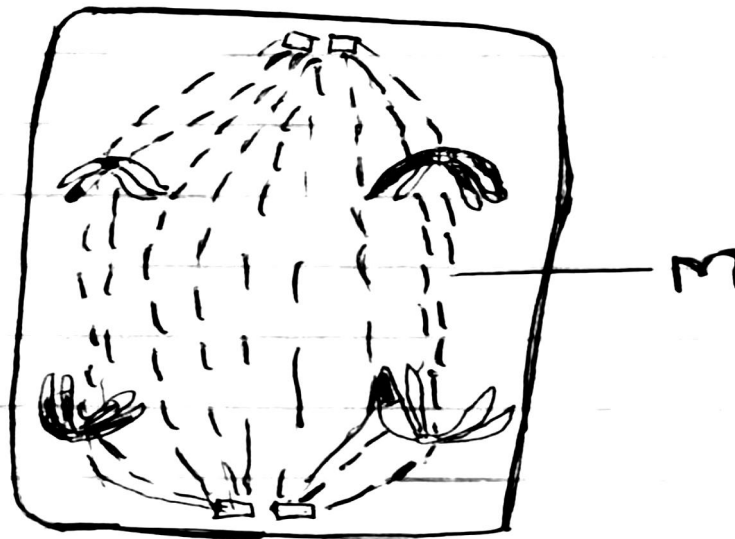
(b) Give two adaptations of ileum to its functions. (2mks)

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14. The diagram below represents a stage during cell division.



(a) (i) Identify the stage of cell division. (1mk)

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(ii) Give two reasons for your answer to (a) (i) above. (2mks)

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(b) Name the structure labelled M. (1mk)

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15. Explain why amoeba cannot burst when placed in hypertonic solution. (2mks)

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(a) Name the organelle that is involved in each manufacture of Lipids. (1mk)

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(b) State three functions of Golgi apparatus. (3mks)

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16. Give the functions of the following parts of human eye:

(a) Lens (1mk)

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(b) Ciliary body. (1mk)

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(c) Cornea (1mk)

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17. A shoot of seedling exposed to light on one side bends towards the source of light as it grows.

(a) Name the response exhibited by the shoot of the seedling. (1mk)

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(b) Explain how the bending towards the source of light occurs. (3mks)

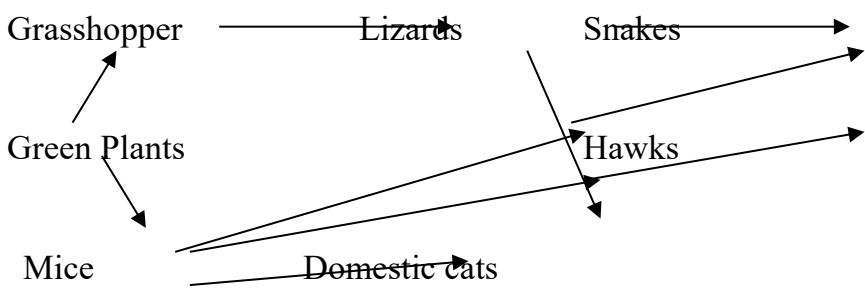
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18. The chart below show s a feeding relationship in a certain eco-system.



(a) Construct two food chains ending with a tertiary consumer in each case. (2mks)

(b) Name one secondary consumers in the food web. (1mk)

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19. State the functions of the following parts of a nephron.

(i) Loop of henle (1mk)

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(ii) Distal convoluted tubule (1mk)

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20. A flower was found to have the following characteristics:

- Inconspicuous petals
- Long feathery stigma
- Small, light pollen grains

(a) What is the likely agent of pollination of the flower? (1mk)

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(b) What is the significance of the long feathery stigma in the flower? (1mk)

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21. Explain how the following factors determine the daily energy requirement in humans.

(a) Age (1mk)

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(b) Occupation (1mk)

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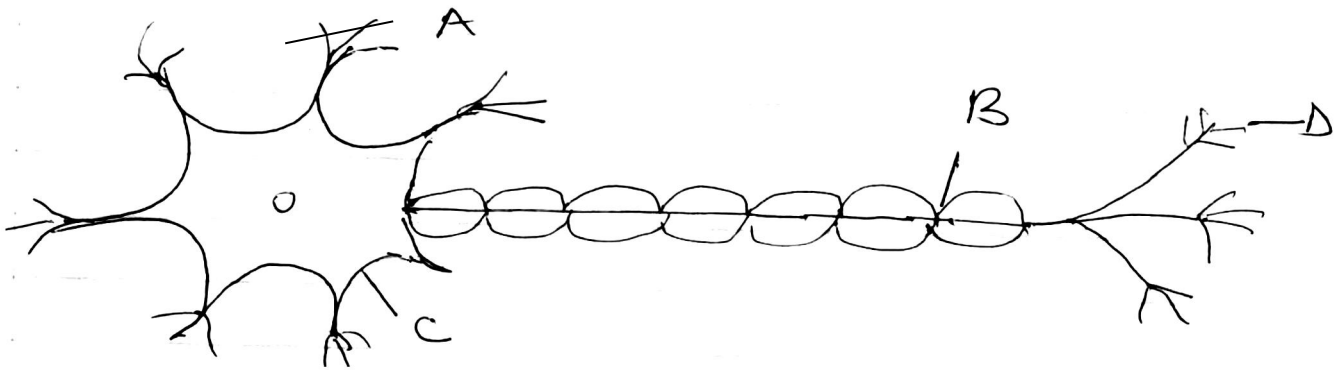
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(c) Sex (1mk)

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22. Study the diagram below and answer the questions that follow.



(a) Name the parts labelled:- (3mks)

A:

C:

D:

(b) State the function of part labelled B. (1mk)

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23. Most terrestrial plants do not grow well in water-logged soils. Give a reason for this. (1mk)

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24. State the mode of a sexual reproduction exhibited by the following organisms:

- Yeast (1mk)

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- Mushroom (1mk)

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25. Give reasons for each of the following:

(a) Constant body temperature is maintained in mammals. (2mks)

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(b) Low blood sugar level is harmful to the body. (1mk)

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26. (a) Explain what is meant by a test-cross as used in genetics. (1mk)

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(b) Determine the probability of a couple with blood group AB getting a child with blood group B.
(Show your working).

27. Name the end products of the light stage of photosynthesis. (2mks)

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